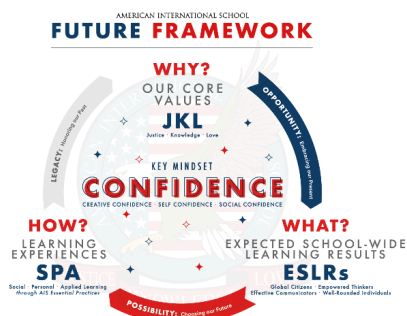




KEY MINDSET CONFIDENCE

LEGACY · Honoring our Past
OPPORTUNITY · Embracing our Present
POSSIBILITY · Choosing our Future

KEY TEACHING & LEARNING EXPERIENCES AT AIS SOCIAL · PERSONAL · APPLIED

SCHOOL MISSION:
To provide a comprehensive and challenging academic program that develops life-long learners and prepares students to meet the responsibilities and expectations of global citizenship.

OUR PURPOSE:
To create students who are Global Citizens, Empowered Thinkers, Effective Communicators, and Well-Rounded Individuals.

AIS WHOLE SCHOOL NORMS

- AIS Core Values drive decisions and actions
- Ensure all voices are heard
- Assume best intentions
- Always go to the source
- Start and end on time
- Be fully present
- Tough on ideas and soft on people
- We are a caring community

OUR CORE VALUES: **Justice** **Knowledge** **Love**

GEOMETRY

Unit 4: Congruence & Similarity Final Assessment (Part I)

Full Name:	Grade:	Date:
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Learning Standards:
 HS.M.AG.C.1 - Use clear and logical justification when solving problems and explaining your understanding.
 HS.M.G.1.G.CO.C.9 - G.CO.C.9 Prove theorems about lines and angles [2-5]
 HS.M.G.1.N.Q.A.1 - Use units as a way to understand problems and to guide the solution of multi-step problems; choose and interpret units consistently in of multi-step problems; choose and interpret units consistently in formulas; choose and interpret the scale and the origin in graphs and data displays.
 HS.M.G.1.HSG.CO.C.10 - HSG.CO.C.10 Prove theorems about triangles. [2,3]
 HS.M.G.2.G.CO.B.7 - G.CO.B.7 Use the definition of congruence in terms of rigid motions to show that two triangles are congruent if and only if corresponding pairs of sides and corresponding pairs of angles are congruent. [4,5]
 HS.M.G.3.G.CO.C.10 - G.CO.C.10 Prove theorems about triangles. [2,3]
 HS.M.G.3.G.GPE.B.4 - G.GPE.B.4 Use coordinates to prove simple geometric theorems algebraically [1]

INSUFFICIENT EVIDENCE	BEGINNING PROFICIENCY	APPROACHING PROFICIENCY	DEMONSTRATING PROFICIENCY	MASTERING PROFICIENCY
I am unable to solve basic problems. I can never apply this standard when solving problems. I am inaccurate when explaining my thinking.	I am beginning to solve basic problems. I can rarely apply this standard when solving problems. I am minimally accurate when explaining my thinking.	I can solve rudimentary problems including those involving simple scenarios. I can sometimes apply this standard when solving problems. I can somewhat accurately explain my thinking.	I can solve familiar problems including those involving real-world scenarios. I can usually apply this standard when solving problems. I am generally accurate when explaining my thinking.	I can solve unfamiliar problems, and analyze unique solutions. I can consistently apply this standard when solving problems. I can accurately and coherently explain my thinking. I can make connections between this concept and other areas of mathematics.

Direction:

- Solve each of the following problems.
- Do not spend too much time on one problem.

01

Use the coordinates below to determine if $\triangle ABC$ and $\triangle DEF$ are congruent.

$$\triangle ABC : A(2, -8), B(-5, -2), C(-7, 3); \triangle DEF : D(-9, 7), E(-11, 12), F(-2, 1)$$

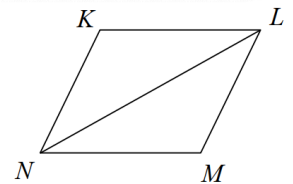
$\overline{AB} =$	$\overline{DE} =$
$\overline{BC} =$	$\overline{EF} =$
$\overline{AC} =$	$\overline{DF} =$

Are the triangles congruent? If yes, explain your reasoning and write a congruency statement.

02

Given: $\overline{NL}$ bisects $\angle KNM$ and $\angle KLM$.

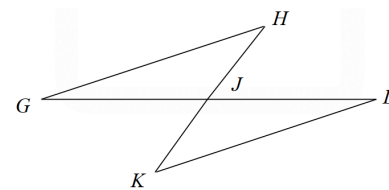
Prove: $\triangle NKL \cong \triangle NML$



Statements	Reasons

03

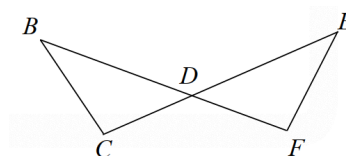
Given: $\overline{GH} \parallel \overline{KL}$, J is the midpoint of $\overline{HK}$
 Prove: $\triangle GHJ \cong \triangle LKJ$



Statements	Reasons

04

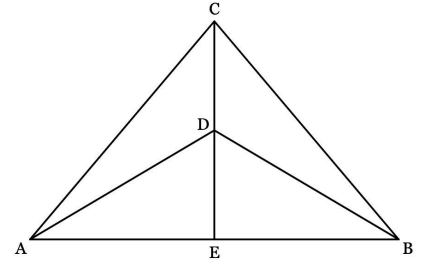
Given: D is the midpoint of $\overline{BF}$ and $\overline{CE}$
 Prove: $\overline{BC} \cong \overline{EF}$



Statements	Reasons

Given: $\triangle ADE \cong \triangle BDE$

Prove: $\angle ACD \cong \angle BCD$



Statements	Reasons

END OF THE ASSESSMENT